

The Proximate Determinants of Fertility

John Bongaarts

ABSTRACT. *The proximate determinants of fertility are the biological and behavioral factors through which socioeconomic and environmental variables operate to influence the rate of childbearing in a population. Three biological proximate variables are identified: permanent sterility, the probability of conception, and intrauterine mortality. Although these variables lower the actual rate of reproduction to far below its potential maximum, they contribute little to trends in fertility. Changes over time in fertility are largely due to changes in the behavioral proximate determinants. The most important of these is the prevalence of contraceptive use, but the duration of breastfeeding, the practice of induced abortion, patterns of union formation, and contraceptive effectiveness are or have been major determinants of levels and trends in fertility in most societies.*

In analyzing the factors that influence levels and trends in fertility, demographers often distinguish between two general types of determinants: (1) proximate variables and (2) socioeconomic and environmental "background" variables.¹ The latter include social, cultural, economic, institutional, psychological, health and environmental factors. The proximate determinants, on the other hand, consist of the biological and behavioral factors through which the background variables must operate to affect fertility.

Background variables → Proximate determinants → Fertility
(e.g., education) (e.g., contraception)

The principal characteristic of a proximate determinant is its direct influence on fertility. In contrast, socioeconomic variables can affect fertility only indirectly by modifying the proximate determinants. As a consequence, fertility differentials among populations or trends in fertility over time can always be traced to variations in one or more of the proximate variables.

In the subsequent discussion, the terms "fertility" and "fecundity" will be encountered repeatedly. Since they are not always used consistently in the demographic and

John Bongaarts is Senior Associate at the Center for Policy Studies, The Population Council, and Associate Professor (adjunct), Department of Population and Family Health, Columbia University. He holds an M.S. in systems analysis from the Eindhoven Institute of Technology, The Netherlands, and a Ph.D. in physiology and biomedical engineering from the University of Illinois, and was a Post-Doctoral Fellow in population dynamics at the Johns Hopkins University. He is the author of four books and numerous articles on population studies.

biomedical literature, a brief definition will be provided, following the terminology in the *Multilingual Demographic Dictionary of the United Nations*.² *Fertility* refers to actual reproduction, whereas *fecundity* denotes the biological ability to reproduce. A woman who is bearing offspring is fertile; a woman is considered fecund if she is able to bear children. The opposite terms are, respectively, *infertile* and *infecund*. Infecund is synonymous with sterile. Infecundity always implies infertility, but the reverse is not necessarily true because a fecund woman may choose to remain infertile by not marrying or by using highly effective contraception. The terms fertility and fecundity are also frequently used to refer to actual and potential reproduction in populations. In the following discussion of the proximate determinants of fertility and fecundity, it will be demonstrated that actual rates of reproduction in human populations (fertility) are far below their potential maximum (fecundity) because of the fertility-inhibiting effects of a number of reproductive behaviors.

The Biological Proximate Determinants of Fecundity

A simple quantitative indicator of the fecundity of a population would be the average number of births women would have over their lifetimes in the absence of behavioral constraints on the rate of reproduction. Imagine a population in which all women marry at the age of fifteen and have regular intercourse throughout their reproductive years. Assume further that no contraception or induced abortion is used to control fertility and that breastfeeding (which causes temporary infecundity) is not practiced. What would be the level of fertility in such a population. Since a full-term pregnancy takes only nine months, women could theoretically bear children at the rate of one per year. This would yield a total of thirty-five births per woman over the reproductive years from fifteen to fifty. Needless to say, no population has ever come close to this upper limit of fertility. Behavioral constraints which will be discussed in later sections are one reason for this, but biological constraints are also crucial. The following three biological (proximate) factors reduce fecundity to well below its hypothetical maximum of thirty-five births per woman.

Involuntary Sterility

Menopause signals the definite termination of the potential reproductive years for women. In most populations for which reliable measures are available, the mean age at menopause is near or slightly below fifty years. This suggests that the potential reproductive years extend into the late forties. Although a woman can be certain that she is infecund after menopause, actual sterility or severe subfecundity can set in years earlier. Possible causes of this involuntary sterility include abnormalities of the reproductive system in otherwise healthy women or their sexual partners, and a rise in intrauterine mortality or incidence of anovulatory cycles as women age. The overall incidence of sterility can be estimated with special demographic techniques. The following table presents a standard age pattern of sterility calculated by Henry, as shown in Table 1.³

According to this schedule, approximately 3% of couples are never able to repro-

TABLE 1. Standard Age Pattern of Sterility

Age of Wife	Couples Sterile
20	3 %
25	5 %
30	8 %
35	15 %
40	32 %
45	69 %
50	100 %

duce (primary sterility). With increasing age the proportion who are sterile grows, reaching 100% at age fifty. Women can become sterile at any time between ages of fifteen and fifty, which are normally considered the potential childbearing years. As a consequence of this early onset of sterility, the highest measured mean age at last birth in actual populations is only about forty years, which is well below the mean age at menopause. Clearly, then, involuntary sterility results in a substantive shortening of the potential reproductive years.

Difficulty in Achieving Conception

Conception can only take place during a few days around the time of ovulation in the middle of a woman's menstrual cycle. But even if intercourse takes place during this so-called "fertile period," conception is not assured. (As is customary in the demographic literature, a conception is defined as a fertilization that results in implantation and survives approximately two weeks to be recognizable by a missed menstruation.) Table 2 below gives estimates of the probability of conception fol-

TABLE 2. Probability of Conception
on Different Days

Time of Intercourse	Daily Probability of Conception	
Days before	-6	.04
temperature	-5	.20
shift	-4	.26
	-3	.15
	-2	.27
	-1	.15
Days after	+1	.07
temperature shift		

lowing intercourse on different days before and after the time of ovulation, as measured by a shift in body temperature.⁴

The probability of conceiving on any of these days around the time of ovulation are substantially less than 1.0, reaching a maximum of only about one in four on days two and four before the temperature shift (the relatively low value for Day -3 is in part caused by sampling errors).

The probability of achieving conception can be increased by having intercourse more than once during the fertile period, but even with higher rates of intercourse, this probability reaches only about 0.4 per menstrual cycle.⁵ The most important reason for this reduced chance of conception is a very high risk of embryo mortality. Apparently, only about half of all fertilized ova survive the first two weeks after fertilization. As a consequence of this difficulty in achieving conception, the average waiting time for conception is more than six months in populations in which frequency of intercourse is not deliberately elevated above typical rates of around twice a week. A prolonged conception delay, in turn, lengthens birth intervals and lowers potential fertility.

Spontaneous Intrauterine Mortality

Approximately one in five recognizable conceptions fails to end in a live birth. The risk of intrauterine mortality is highest during the first month after the missed menses following fertilization, when a high proportion of fetuses with genetic abnormalities are expelled. Apparently, embryonic and intrauterine mortality effectively remove defective embryos and fetuses so that nearly all live births are free of abnormalities.

Fecundity

It is difficult to measure a population's fecundity directly, because there are no societies in which behavioral reproductive constraints are absent. Indirect estimates with mathematical models, however, indicate that the above three biological factors (*i.e.*, involuntary sterility, difficulty in achieving conception, and intrauterine mortality) reduce the potential fertility of populations from its theoretical level of thirty-five to only about fifteen births per woman.⁶ This level of fertility will be referred to below as the "fecundity rate."

The Proximate Determinants of Natural Fertility

Natural fertility is defined as fertility in the absence of deliberate attempts to limit family size. In practice, fertility may be considered natural in the absence of contraception or induced abortion, because they are the principal means by which couples control their family sizes.⁷ Natural fertility prevailed in the developed world before the onset of the fertility transition in the second half of the 19th century,

and virtually all countries in the Third World had natural fertility up to the middle of the 20th century.

In many cases, particularly in subSaharan Africa, fertility has remained close to natural until today. Drawing on a variety of statistical sources, demographers have made estimates of rates of reproduction in these populations. Typical levels of natural fertility in Third World countries range from six to seven births per woman with an occasional population reaching as high as eight (the world record-holder is Yemen A. R., with 8.5 births per woman). In historical European populations, natural fertility levels were generally slightly lower, but they rarely fell below five.

These natural fertility rates are obviously far below the fecundity rate of around fifteen births per woman estimated previously. The explanation for this large discrepancy between biologically feasible and actual natural fertility is found in the following behavioral proximate factors.⁸

Breastfeeding

In traditional societies, virtually all infants are given breastmilk for at least the first several months of their lives and, for many infants, breastfeeding is continued until the mother becomes pregnant again. Although its main function is the provision of nutrition to infants, breastfeeding also has an important effect on the population's fertility, because it delays the return of ovulation after each birth. The duration of this lactational anovulation depends on the intensity and duration of breastfeeding. Periods of anovulation of $1\frac{1}{2}$ to two years are common among women who practice prolonged breastfeeding.

The physiological mechanism through which breastfeeding influences ovulation is not yet completely understood, but the key processes involved appear to operate as follows.⁹ The suckling by the infant stimulates receptors in the breast nipple that initiate a neural signal to the hypothalamus, a nerve center in the brain. In response, the discharge of hypothalamic gonadotropic-releasing hormone is suppressed, which results in a decreased secretion of luteinizing hormone and the failure of ovulation. This physiological process identifies frequent nipple stimulation as the critical factor in maintaining anovulation. Only frequent suckling, as during full breastfeeding, produces a sufficient reduction in the production of luteinizing hormone. As the frequency and intensity of breastfeeding decrease (*e.g.*, with the introduction of supplements or bottle feeding), ovulation eventually resumes.

This period of anovulation ends for most women while they are still breastfeeding (except in societies where the breastfeeding interval is very brief). Before the first postpartum ovulation, the chance of conception is zero, while after the resumption of regular ovulation and menstruation, women can conceive again, even if breastfeeding is maintained. If ovulation could be detected readily, a woman could rely on lactational anovulation as a perfect—but short-term—contraceptive. A simple but not entirely reliable indicator of the return of ovulation is the resumption of menstruation. Since ovulation sometimes precedes the first postpartum menstruation, a small proportion of women (5–10%) become pregnant during lactational

amenorrhea. Although this is not a negligible proportion, it is worth noting that the annual failure rates of a number of contraceptive methods are higher.

Abstinence from Sexual Relations Among Breastfeeding Women

The resumption of sexual relations is delayed by at least a few weeks after birth in most populations. In some societies, however, abstinence is much longer. In parts of tropical Africa, couples believe that "semen poisons the milk," and abstinence can last until the child is weaned at age two or three. In such cases, abstinence increases the birth interval and reduces fertility if the abstinence interval exceeds the duration of anovulation.

Delayed Marriage

To be exposed to the possibility of childbearing throughout the potential reproductive period, a woman would have to marry (or enter into a socially sanctioned union) at menarche. There is a substantial delay between menarche and marriage in most traditional societies. This lost reproductive time can have a significant effect on fertility. For example, if menarche occurs at age fourteen or fifteen and marriage at eighteen, then perhaps one birth has been lost over the woman's lifetime, due to this delay. A much larger effect of delayed marriage is found in historical European populations, where age at marriage was often in the mid-twenties.

Marital Disruption

The termination of marriage by either divorce or widowhood occurs in all populations. The demographic significance of these events is variable, in part because the divorce risk is very culture-specific and in part because the probability of remarriage differs widely among societies. The only generalization that can be made with confidence is that the risk of widowhood is closely related to the overall mortality of the population.

Patterns of Reproduction

Figure 1 presents a summary of the joint effects of biological and behavioral proximate determinants on the timing of reproductive events in populations with natural fertility. The first bar in this figure represents a typical reproductive pattern in a traditional developing country. In this example, the mean age at marriage is set at nineteen, and, with a mean birth interval of three years, women have—on average—seven births by age forty before sterility sets in. The three-year birth interval could consist of three components: pregnancy (nine months), postpartum anovulation (eighteen months), and waiting time to a conception that ends in a live birth (nine months).

The second bar depicts the reproductive pattern in a not-atypical historical European society. A mean age at marriage of twenty-five might be combined with an average birth interval of 2.5 years to yield a total of six births per woman. The key

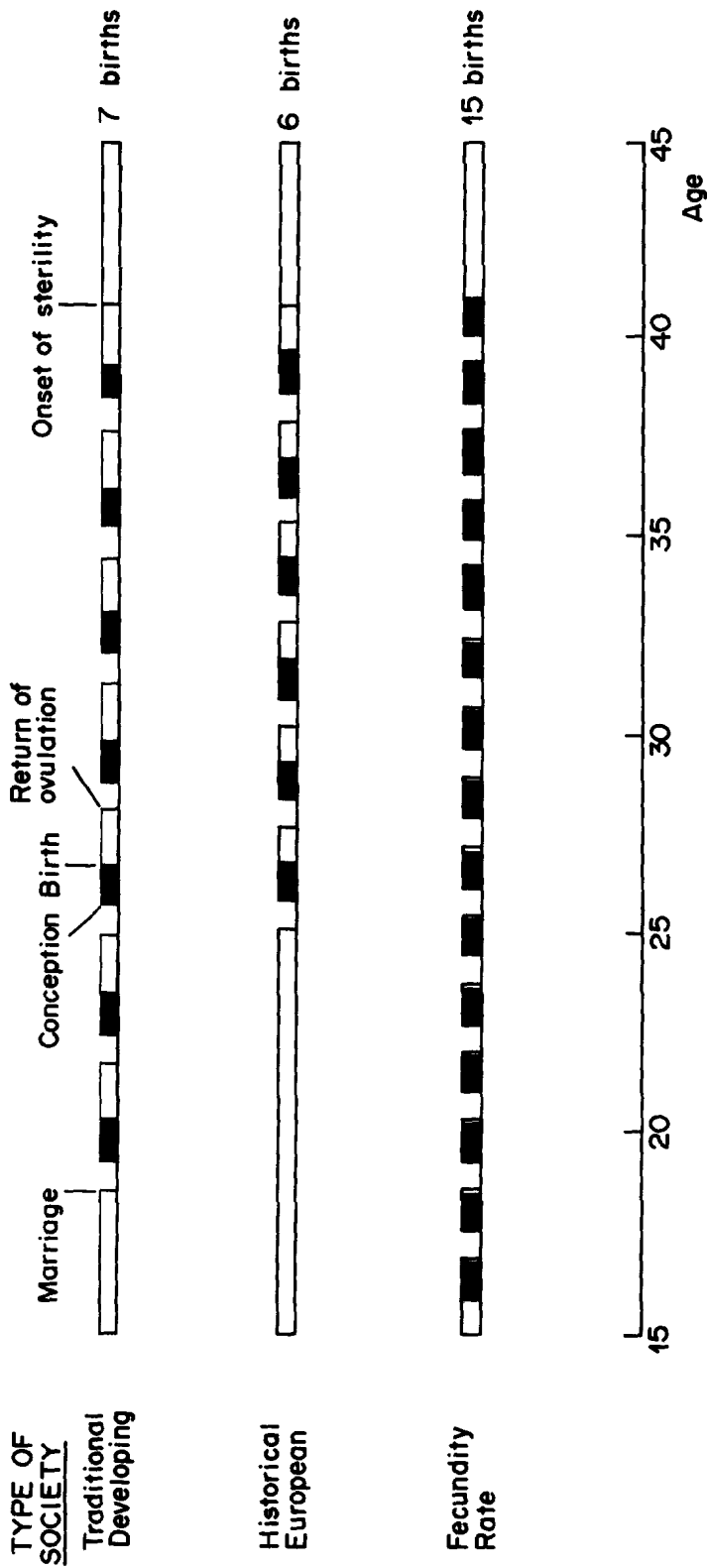


FIGURE 1. Average Timing of Reproductive Events in Selected Types of Societies

difference from the reproductive pattern in developing countries is the much later age at marriage and the shorter birth interval (due to shorter breastfeeding). For comparison, the last bar in Figure 1 gives the timing of events in a hypothetical population without the constraints of marriage, marital disruption, breastfeeding, and postpartum abstinence. In this hypothetical case, birth intervals would average about twenty months, thus yielding a fecundity rate with a total of fifteen births between ages fifteen and forty.

The Proximate Determinants of Controlled Fertility

In the majority of contemporary societies, fertility has declined below the natural levels that prevailed before the onset of the fertility transition. In these populations, couples apply deliberate reproductive control to achieve specific family size objectives. The principal means of deliberate control are contraception and induced abortion.

Contraception

In 1980, approximately 45% of married couples worldwide were current users of contraception. The distribution of methods practiced is estimated to have been as shown in Table 3.¹⁰

Sterilization is the most popular method, followed by the IUD and hormonal contraception. Together these modern methods accounted for three-quarters of all use worldwide in 1980. By definition, contraceptive users intend to eliminate the probability of conceiving. In practice, however, none of the methods other than sterilization reduces this probability to zero. Contraceptive failures can be due either to inappropriate application of the method (*e.g.*, forgetting to take a Pill daily) or to imperfection in the method (*e.g.*, the rhythm method). Failure rates vary widely for different methods, as shown in Table 4 for the US.¹¹

Since these estimates are obtained in a highly developed and well educated society, it is safe to infer that failure rates are substantially higher in most developing

TABLE 3. Methods of Contraception Used

Method	Users
Sterilization	32%
IUD	26%
Pills and injectables	17%
Condoms	9%
Other	<u>15%</u>
All	100%

TABLE 4. Failure Rates for Contraceptive Methods

Method	Users Failing in One Year
Sterilization	0%
Pill	2%
IUD	4%
Condom	10%
Diaphragm	13%
Foam/cream/jelly	15%
Rhythm	19%

countries (except for the IUD and sterilization, which are not susceptible to user errors). The implication of these figures is that—in countries where contraceptive use is widespread—a large proportion of couples experience at least one contraceptive failure and a substantial percentage of all births each year are conceived by contraceptive couples. For example, in Costa Rica, fully 35% of recent births were accidentally conceived, according to respondents in the 1976 World Fertility Survey.¹²

Induced Abortion

Induced abortion is an effective method of birth control. It is also a safe procedure if carried out during the first trimester of pregnancy and under hygienic conditions by competent medical personnel. Induced abortion alone, however, is not a desirable method of birth control, since an abortion averts on average less than one birth. This is the case in part because some pregnancies terminated by induced abortion would have ended in a spontaneous intrauterine death, and in part because, after an induced abortion, a woman returns to the ovulatory state sooner than she would have if the pregnancy had ended in a live birth followed by a postpartum anovulation interval.

For example, a woman who would have an average birth interval of three years under natural conditions would need to have induced abortions at intervals of about fourteen months (*e.g.*, nine months' waiting time to conception plus three months' gestation before the abortion and two months of postpartum anovulation) if she wanted to avoid having live births by relying solely on abortion. As a consequence, she would have to have ten abortions if she wanted to eliminate childbearing for a twelve-year period. Since she would have had only four births during this same period in the absence of fertility control, each abortion averts only 0.4 births (four-tenths). It is for this reason that induced abortion is best used as a backup in case of contraceptive failure. When practiced in combination with contraception, each abortion averts about 0.8 births, and the number of abortions required to avert all births over a given number of years is only a small fraction of what it would be without contraception.

Trends in the Proximate Variables During the Fertility Transition

Reproductive behavior changes drastically during the development process that transforms a traditional agricultural society into a modern industrial one. Over the course of this transition, fertility declines from natural levels of around six or seven births per woman to near or even below the replacement levels of two children per family. The large increase in the practice of contraception is the most important proximate cause of this trend, but other proximate variables change also.

The following discussion of trends in proximate determinants will focus on broad patterns of recent reproductive change in the Third World, because this is a subject of interest to policymakers and because more is known about contemporary than about historical populations. As Table 5 indicates, rapid changes in fertility have taken place in recent decades, particularly in the 1970s. During the 1950s, fertility was high and virtually stable in all but a small proportion of countries in the Third World. At that time, fertility was close to natural (the small temporary dip in fertility in East Asia around 1960 was due to a severe famine in China). During the 1960s, fertility dropped only in Latin America, but the 1970s saw an unprecedentedly rapid change in the fertility of China and a less rapid yet substantial decline in South Asia. Despite the absence of a significant trend in fertility in Africa, the Third World as a whole experienced a drop in fertility from above six in the early 1960s to near four births per woman in the early 1980s. Attention now turns to an examination of the trends in the proximate variables that accompany these fertility transitions.

The Biological Determinants of Fertility

Although a lack of accurate measures prevents a conclusive assessment, the available evidence suggests that there are, in general, only small differences in sterility, probability of conception, and intrauterine mortality between traditional and modern societies. One could hypothesize that improvements in nutrition and health con-

TABLE 5. Fertility Trends (Average Number of Births per Woman) from 1950 to 1985 in Major Regions of the Developing World

	Year					
	1950-55	1960-65	1965-70	1970-75	1975-80	1980-85
Africa	6.5	6.6	6.6	6.5	6.4	6.3
Latin America	5.9	5.9	5.5	5.0	4.5	4.1
East Asia ^a	6.2	5.9	6.0	4.7	2.9	2.4
South Asia	6.0	6.1	6.0	5.7	5.1	4.6
All developing countries	6.1	6.1	6.0	5.4	4.5	4.1

^aExcluding Japan.

Source: United Nations (see Note 13).

ditions reduce sterility, but these effects don't amount to much.¹⁴ The mean age at last birth in well nourished and health populations with natural fertility is very similar to that in their poorer counterparts, which suggests that the onset of sterility does not vary much with improved standards of living. (One exception to this generalization should be noted: Any rise in venereal disease-induced sterility would, of course, affect fertility. Substantial levels of pathological sterility, however, are at present only found in selected populations in tropical Africa.)

Similarly, available estimates of spontaneous intrauterine mortality show no clear differentials between developed and developing societies. This is the case despite the statistically significant influence of health and nutritional status on the stillbirth rate. This finding is explained by the fact that stillbirths make up only a small part of all fetal mortality, and intrauterine mortality in the early months of pregnancy is apparently only minimally affected by environmental factors. Finally, concerning the probability of conception, there is no evidence that the viability of ova and sperm or embryonic mortality are significantly affected by chronic moderate malnutrition or general poor health. (Extreme deprivation, however, as during famines, does reduce fecundity.) As noted earlier, the probability of conception also depends on the frequency of intercourse. This is one variable for which reliable trend data are nonexistent. Nevertheless, a lack of consistent differentials in waiting times to conception among countries at different levels of development suggests that there are no major trends in frequency of intercourse during the fertility transition.

Breastfeeding

Longitudinal measurements of the duration of breastfeeding or lactational amenorrhea are extremely rare, and generalizations about changes in breastfeeding behavior over time must therefore rely largely on comparisons of societies at different levels of socioeconomic development. Table 6 gives the average duration of breastfeeding for 38 developing countries that participated in the World Fertility Survey (WFS). (These surveys were conducted in the years from 1974 to 1982 with the majority taking place in the late 1970s.)

Durations of breastfeeding of a year and a half or more were observed in the most traditional societies, including all but one of the subSaharan WFS countries, as well as Bangladesh, Nepal, Pakistan and Indonesia. On the other hand, average breastfeeding intervals of nine months or less were found in the more modern WFS countries, such as Colombia, Costa Rica, Dominican Republic, Jamaica, Malaysia, Mexico, Panama and Venezuela. In contemporary developed countries in North America and Europe, breastfeeding is still shorter—typically no longer than a few months. These data indicate that, with development, breastfeeding duration declines, thus reducing the fertility-inhibiting effect of lactational amenorrhea, as well as any breastfeeding related abstinence. Factors that tend to encourage this change in breastfeeding are the availability of hygienic and nutritionally adequate food supplements, improvements in health conditions which make breastfeeding less of a requirement for child survival, and changes in women's labor-force participation that make it difficult to continue time-consuming breastfeeding.

TABLE 6. Average Duration of Breastfeeding in Selected Developing Countries with WFS Surveys

Duration of Breastfeeding (months)	Country
30 -	
-	Bangladesh
-	
-	
25 -	Nepal
-	Indonesia
-	
-	Sri Lanka
20 -	Lesotho
-	Benin, Senegal, Pakistan, Thailand
-	Cameroon, Ghana, Ivory Coast
-	
-	Kenya, Egypt, Mauritania, Sudan, Korea
15 -	Haiti
-	Morocco, Tunisia
-	Philippines, Peru
-	Syria, Ecuador
-	Jordan, Yemen, Paraguay
10 -	Fiji
-	Colombia, Dominican Republic, Mexico
-	Jamaica, Trinidad, Tobago
-	Venezuela, Panama, Guyana
-	Malaysia
5 -	Costa Rica
-	
-	
-	
0 -	

Source: J. Cleland and J. Hobcraft, *Reproductive Change in Developing Countries* (Oxford: Oxford University Press, 1985) p. 15.

Marriage

With socioeconomic development and the accompanying changes in roles of women, the timing of marriage also changes. Findings from the World Fertility Survey summarized in Table 7 provide estimates of age at first marriage. The average age at first marriage in these forty WFS countries ranges from sixteen years in Bangladesh

TABLE 7. Average Age at Marriage in Selected Developing Countries with WFS Surveys

Average Age at Marriage	Country
25 -	Sri Lanka, Philippines
-	Tunisia
-	Korea, Malaysia, Thailand, Peru, Costa Rica
-	Jordan, Syria, Fiji, Colombia, Ecuador, Paraguay, Venezuela, Mexico, Haiti
-	Egypt, Morocco, Jordan, Turkey, Dominican Republic, Panama
20 -	Lesotho, Pakistan, Guyana, Trinidad and Tobago
-	Ghana, Kenya, Nigeria, Mauritania, Indonesia
-	Benin, Cameroon, Ivory Coast, Senegal, Jamaica
-	Yemen, Nepal
-	Bangladesh
15 -	

Source: J. Cleland and J. Hobcraft, *Reproductive Change in Developing Countries* (Oxford: Oxford University Press, 1985), p. 16.

to twenty-five years in Sri Lanka and the Philippines. Marriage tends to occur earlier in the more traditional societies, which suggests that age at marriage rises over time as a society moves through the fertility transition. Direct confirmation of this conclusion is available from WFS, because the mean age at marriage at the time of the survey can be compared with the mean age at marriage reported by the older women who married two or three decades before the survey data.¹⁷ The results (not shown here) indicate that the mean age at marriage rose in the large majority of the WFS countries for which this comparison over time could be made. The average increase was 2.2 years, while in a number of countries (*e.g.*, Korea, Malaysia, Sri Lanka) the age of marriage rose as much as five years.

A comprehensive discussion of nuptiality-related proximate variables should also include a comment on marital disruption (through widowhood, separation or divorce), remarriage, and permanent celibacy. This subject, however, will not be addressed further here in part because it is highly complex and has not been studied intensively by demographers, and in part because variations in these variables play only a minor role in determining levels and trends in fertility.

Contraception and Induced Abortion

Because government efforts to reduce fertility in the developing world have been focused on encouraging wider practice of contraception, more information is available about levels and trends in this proximate variable than for the others. The principal measure of the extent of contraceptive use in a population is the so-called prevalence rate, which equals the proportion of married women of reproductive age

that practices a method of contraception at a given point in time (male methods are included). Sources of prevalence estimates include population, fertility or prevalence surveys and, in a few cases, censuses. Program service statistics are another source of contraceptive data in countries with large and comprehensive family planning programs. Even in populations, however, where the family planning program is virtually the only supplier of contraceptives early in the transition, private sources of supply become increasingly more important over time. As a consequence, service statistics provide generally less reliable estimates of overall prevalence than surveys.

Table 8 presents prevalence levels and trends for four major regions of the Third World. For all developing countries, combined contraceptive prevalence rose from 9% in 1960–65 to 38% in 1980. In the early 1960s, each of the major regions had low prevalence rates, ranging from an estimated 5% in Africa to 14% in Latin America. Although direct measures are not available for earlier periods, it is likely that prevalence has been low for centuries up to the 1950s, because fertility rates were high and close to natural levels. By 1980, very rapid increases had occurred in East Asia (which is dominated by China) and to a lesser—but still impressive—extent in Latin America. East Asia actually reached a prevalence rate of 69%, which is very similar to levels observed in contemporary developed societies. In Latin America, prevalence reached 43% and a more modest but nevertheless substantial rise in prevalence to 24% occurred in South Asia. Only in Africa did prevalence change only marginally. All these trends in contraceptive use are consistent with declines in fertility over the same period: the more rapid the rise in prevalence, the greater the decline in fertility (see Tables 5 and 8).

The inverse relationship between fertility and prevalence observed in these longitudinal data is also evident in Figure 2, which plots average levels of fertility in groups of countries with different prevalence rates in 1980. Countries with contraceptive prevalence of less than 10% have on average a fertility rate of 6.5 births per woman. Most of the countries in this lowest prevalence category are African. At

**TABLE 8. Contraceptive Prevalence Estimates
by Region 1960–65 to 1980–81**

	Contraceptive Prevalence (%)	
	1960–65	1980–81
Africa	5	11
Latin America	14	43
East Asia ^a	13	69
South Asia	7	24
All developing countries	9	38

^aExcluding Japan.

Source: Bongaarts¹⁹ and United Nations.²⁰

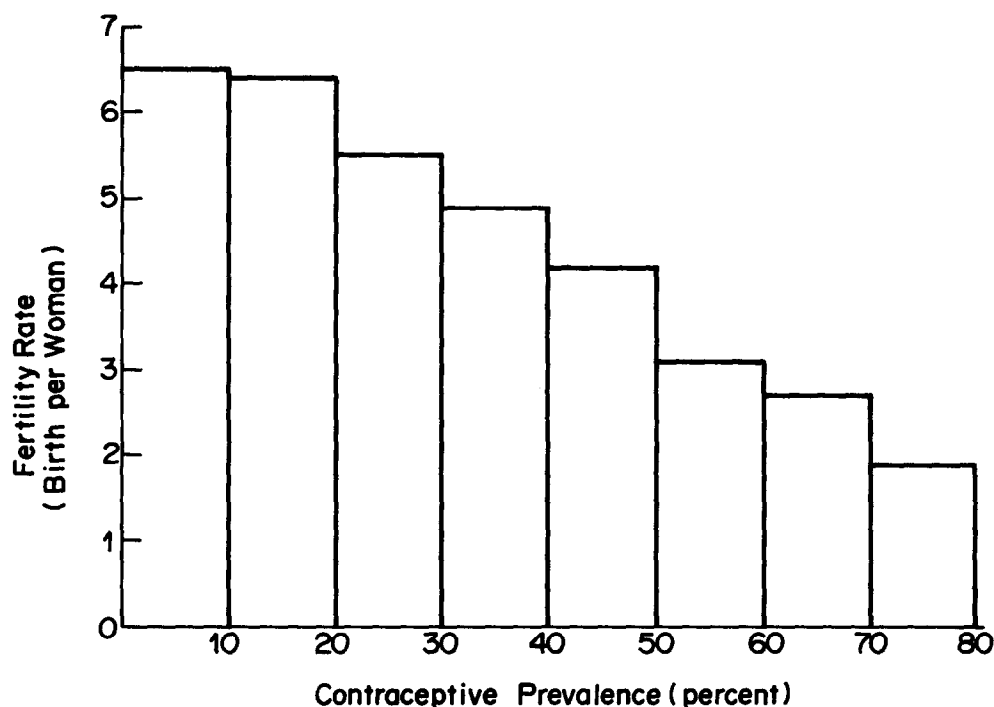


FIGURE 2. Average Fertility Rate in Groups of Countries with Different Levels of Contraceptive Prevalence (ca. 1980).

Source: United Nations²⁰ and Lapham and Mauldin²¹

the other extreme, fertility averaged only 1.9 births per woman in countries in which over 70% of married women (or their husbands) were contracepting. Nearly all countries in this group were developed. Over the course of a full transition from natural to replacement fertility, a population's prevalence level typically rises from a few percent to around 75%.

The fertility impact of contraception is determined not only by the prevalence level, but also by the level of effectiveness, which depends in part on the mix of methods. It is difficult to make generalizations about correlates of method distribution, because it varies greatly among populations.

Table 9 gives the method distribution by major regions of the world in 1980. In Africa, more than half the users relied on the Pill, while in South Asia sterilization was the most popular, and in East Asia the IUD was the most widely used method. An interesting and surprising finding in Table 9 is the sharp contrast between the developed and the developing countries. In the latter, sterilization and the IUD accounted for about three-quarters of all use. In contrast, a majority of women in the developed world relied on "other" (mostly traditional) methods of contraception, and sterilization and the IUD were the least popular methods. It is worth noting, however, that over the past decade, the use of modern methods has been rising relative to other methods in North America and Western Europe.

TABLE 9. Contraceptive Method Distribution for Regions of the World, 1980-81

	Contraceptive Method Used (%)				Total
	Sterilization	IUD	Pills and Injectables	Other	
Africa	8	14	55	22	100
Latin America	24	14	34	28	100
East Asia ^a	36	49	8	6	100
South Asia	54	7	19	19	100
All developing regions	40	33	14	13	100
All developed regions	14	7	23	56	100

^aExcluding Japan.

Source: United Nations (Note 10).

In the absence of reliable abortion statistics in virtually the entire developing world, it is not possible to provide an accurate assessment of levels and trends in induced abortion. Laws that prohibit or severely restrict the use of induced abortion exist in all but a handful of developing countries. In only three of these countries are reliable statistics available: Cuba, Tunisia and Hong Kong. For 1983, induced abortion rates in these three cases are estimated at, respectively, 1.60, 0.41 and 0.32 abortions per woman over her lifetime (average).²³ Other information, ranging from relatively unreliable statistics to educated guesses, suggests that only in China and North Korea are abortion rates of the same order of magnitude as in Cuba. In most of the rest of the Third World, abortion rates are likely to be similar to or even lower than they are in Tunisia and Hong Kong.

Joint Effects of the Proximate Determinants

A change in a population's fertility level is by definition due to the combined effects of all the proximate determinants. The discussion in the preceding section has identified the principal trends in reproductive behavior that occur during fertility transition in the Third World. Typically, breastfeeding and related amenorrhea decline, while age at marriage rises, and the practice of deliberate fertility control through contraception becomes widespread. Of these three trends in reproductive behavior, the last two (marriage and contraception) tend to reduce fertility while the first (breastfeeding) exerts an upward pressure on fertility. In the majority of populations that experience socioeconomic development, the rise in contraceptive prevalence and age at marriage are more than sufficient to compensate for any upward pressure on fertility that arises from the decline in breastfeeding. As a consequence, fertility declines throughout the transition.

But this result is only obtained if these proximate determinants change more or less simultaneously. There are interesting and important exceptions to this pattern. Populations can experience a temporary rise in fertility early in the transition if the

decline in the traditional childspacing practices of breastfeeding and postpartum abstinence occur initially at a more rapid pace than changes in the other proximate variables. This is very likely the reason for the rise of fertility in Kenya, which reached a record-breaking level of 8.1 births per woman in the 1970s. Future increases in contraceptive use will no doubt bring an eventual reduction in fertility, but Kenya's experience is certainly an unusual one. The powerful fertility-enhancing effect of reduced breastfeeding is illustrated by the following simple numerical example: In the typical reproductive pattern for a traditional developing country shown in Figure 1, women had an average of seven births at intervals of three years. Half of these birth intervals were assumed to be spent in lactational amenorrhea. If this amenorrhea interval were to be shortened to, say, six months, then the average birth interval would be reduced to 24 months and fertility would rise to 10.5 births per woman. This positive fertility effect of changes in breastfeeding behavior can clearly play an important role in determining fertility trends during the early phases of the fertility transition. As a consequence, Kenya's experience may well be repeated in other countries, especially in African populations with prolonged breastfeeding and abstinence practices.

Conclusion

The recent gain in the understanding of the proximate determinants has been made possible by an interdisciplinary approach to the study of the human reproductive process. By drawing on insights from two traditionally separate disciplines, biology and demography, it has been possible to make rapid progress in explaining levels, trends and differentials in fertility in terms of the proximate variables. Biological factors are crucial in determining overall fertility levels in human populations, but trends in fertility and differentials among populations are largely due to variations in reproductive behaviors.

This new knowledge has produced tentative explanations for a number of previously puzzling phenomena. An example is the occasional rise in fertility during the earliest phase of the fertility transition. This anomaly can largely be explained by the rapid abandonment of traditional constraints on natural fertility (*e.g.*, lactational anovulation) at a time when the uptake in deliberate fertility control through contraception is still limited. Since socioeconomic variables affect fertility only through the proximate determinants, the initial net fertility impact of development can be positive, negative or negligible, depending on the extent to which fertility-enhancing changes in one set of proximate variables are offset by fertility-reducing effects of another set. A continued rise in deliberate fertility control through contraception or induced abortion, however, will inevitably bring about a reduction in fertility after the first phase of the fertility transition.

Notes

1. This distinction between types of fertility determinants was first made by Kingsley Davis and Judith Blake, "Social Structure and Fertility: An Analytic Framework," *Economic Development and Cultural Change*, Vol. 4 (1956), pp. 211-235. For a recent discussion on the proximate determinants and their quantitative relation-

- ship to fertility, see John Bongaarts and Robert Potter, *Fertility, Biology and Behavior* (New York: Academic Press, 1983).
2. United Nations, "Multilingual Demographic Dictionary," *Population Studies*, no. 29 (New York: United Nations, 1959).
 3. Louis Henry, "French Statistical Research in Natural Fertility," in M. Sheps and J. Ridley, eds., *Public Health and Population Change* (Pittsburgh: University of Pittsburgh Press, 1965).
 4. D. Schwartz, M. MacDonald and V. Heuchel, "Fecundability, Coital Frequency, and the Viability of Ova," *Population Studies*, Vol. 34, no. 2 (1980), pp. 397-400.
 5. J.C. Barrett and J. Marshall, "The Risk of Conception on Different Days of the Menstrual Cycle," *Population Studies*, Vol. 23, no. 2 (1969), pp. 455-461.
 6. J. Bongaarts and R. Potter, *op. cit.*
 7. Louis Henry, "Some Data on Natural Fertility," *Eugenics Quarterly*, Vol. 8, no. 2 (1961), pp. 81-91.
 8. The discussion of this topic is drawn in part from J. Bongaarts, "The Transition in Reproductive Behavior in the Third World," in J. Menken, ed., *World Population and U.S. Policy: The Choices Ahead* (New York: W.W. Norton and Co., 1986).
 9. Roger Short, "Breastfeeding," *Scientific American*, Vol. 250, no. 4 (1984), pp. 35-41.
 10. United Nations, *Recent Levels and Trends of Contraceptive Use, As Assessed in 1983* (New York: United Nations, 1984).
 11. B. Vaughn, J. Trussell, J. Menken and E. Jones, "Contraceptive Failure Among Married Women in the United States," *Family Planning Perspectives*, Vol. 9, no. 6 (1977), pp. 251-258.
 12. N. Goldman and L. Paul, personal communication, special tabulation of WFS data, September 1984.
 13. United Nations, *World Population Prospects: Estimates and Projections As Assessed in 1984* (New York: United Nations, 1986).
 14. R. Gray, "Factors Affecting Natural Fertility Components: Health and Nutrition," in R. Bulatao and R. Lee, eds., *Determinants of Fertility in Developing Countries: A Summary of Knowledge* (New York: Academic Press, 1983).
 15. John Cleland and John Hobcraft, *Reproductive Change in Developing Countries* (Oxford: Oxford University Press, 1985).
 16. *Ibid.*
 17. Peter McDonald, *Nuptiality and Completed Fertility: A Study of Starting, Stopping and Spacing Behavior*. Comparative Studies no. 35 (Voorburg, The Netherlands: International Statistical Institute, 1984).
 18. United Nations, 1984, *op. cit.*, Note 10.
 19. John Bongaarts, "Implications of Future Fertility Trends for Contraceptive Practice," *Population and Development Review*, Vol. 10, no. 1 (1984), pp. 341-351.
 20. United Nations, *World Population Prospects: Estimates and Projections As Assessed in 1982* (New York: United Nations, 1985); United Nations, 1983, *op. cit.*
 21. R. Lapham and W.P. Mauldin, "Contraceptive Prevalence: The Influence of Organized Family Planning Programs," *Studies in Family Planning*, Vol. 16, no.3 (1985), pp. 117-137.
 22. United Nations, 1984, *op. cit.*
 23. C. Tietze and S. Henshaw, *Induced Abortion: A World View* (New York: The Alan Guttmacher Institute, 1986).